

Customer Segmentation Using K-Means Clustering

Synent Technologies Data Science Internship

Task 6: Customer Segmentation

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Problem Statement

Businesses often have thousands of customers with different purchasing behaviors.

The goal of this project is to identify distinct customer groups based on annual income and spending patterns using K-Means Clustering.

These customer segments can help businesses:

- Design targeted marketing campaigns
- Improve customer retention
- Increase revenue
- Understand purchasing behavior

```
In [1]: pip install pandas numpy matplotlib seaborn scikit-learn
```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas in c:\programdata\anaconda3\lib\site-packages (2.3.3)
Requirement already satisfied: numpy in c:\programdata\anaconda3\lib\site-packages (2.3.5)
Requirement already satisfied: matplotlib in c:\programdata\anaconda3\lib\site-packages (3.10.6)
Requirement already satisfied: seaborn in c:\programdata\anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: scikit-learn in c:\programdata\anaconda3\lib\site-packages (1.7.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\programdata\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\programdata\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: scipy>=1.8.0 in c:\programdata\anaconda3\lib\site-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in c:\programdata\anaconda3\lib\site-packages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\programdata\anaconda3\lib\site-packages (from scikit-learn) (3.5.0)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

```
In [2]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

Dataset Details

Dataset Name: Mall Customer Segmentation Dataset

Source: Kaggle

Number of Records: 200 Customers

Number of Features: 5

Features:

1. CustomerID
2. Gender
3. Age
4. Annual Income (k\$)
5. Spending Score (1-100)

Target: Unsupervised Learning Project (No Target Variable)

```
In [3]: df = pd.read_csv("Mall_Customers.csv")  
  
df.head(10)
```

```
Out[3]:
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
5	6	Female	22	17	76
6	7	Female	35	18	6
7	8	Female	23	18	94
8	9	Male	64	19	3
9	10	Female	30	19	72

```
In [5]: df.shape
```

```
Out[5]: (200, 5)
```

```
In [6]: df.columns
```

```
Out[6]: Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',  
              'Spending Score (1-100)'],  
              dtype='object')
```

```
In [7]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CustomerID                           200 non-null    int64
1   Gender                               200 non-null    object
2   Age                                   200 non-null    int64
3   Annual Income (k$)                   200 non-null    int64
4   Spending Score (1-100)               200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB

```

```
In [8]: df.describe()
```

```
Out[8]:
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
In [9]: df.isnull().sum()
```

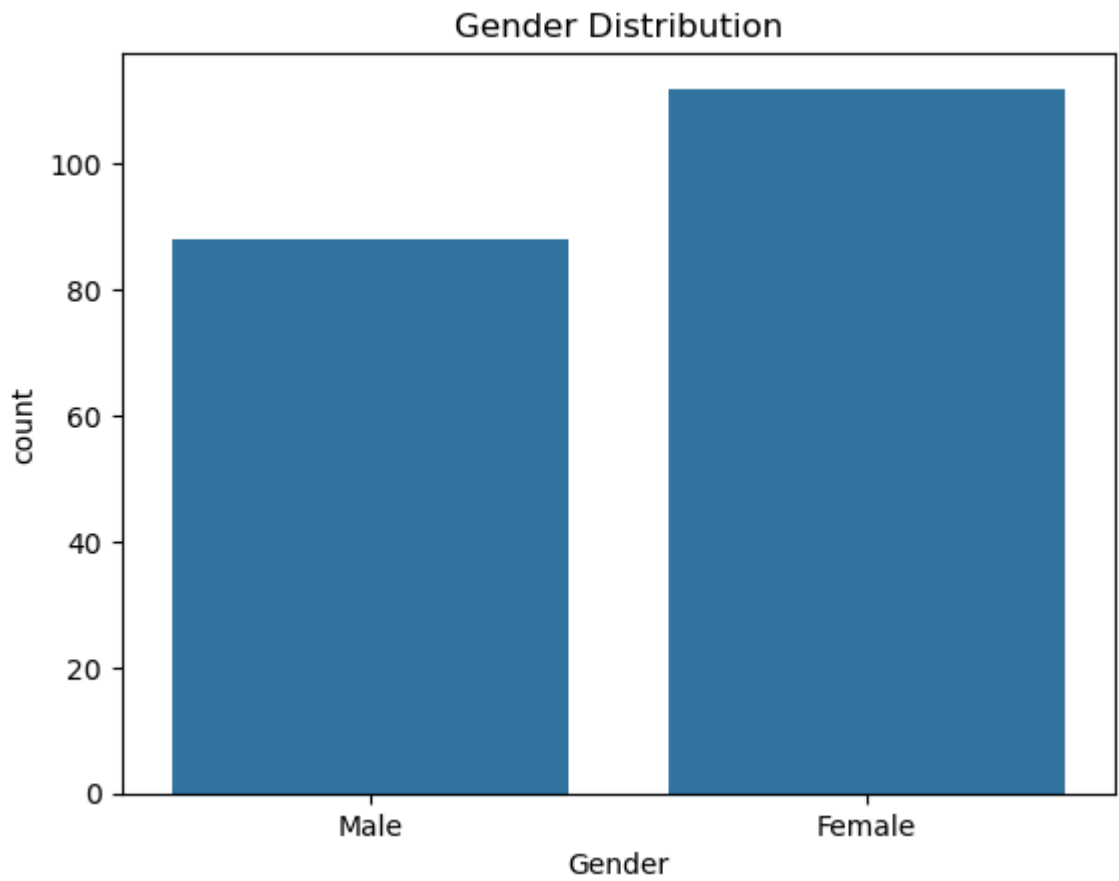
```
Out[9]: CustomerID      0
Gender      0
Age      0
Annual Income (k$)      0
Spending Score (1-100)  0
dtype: int64
```

```
In [10]: df.duplicated().sum()
```

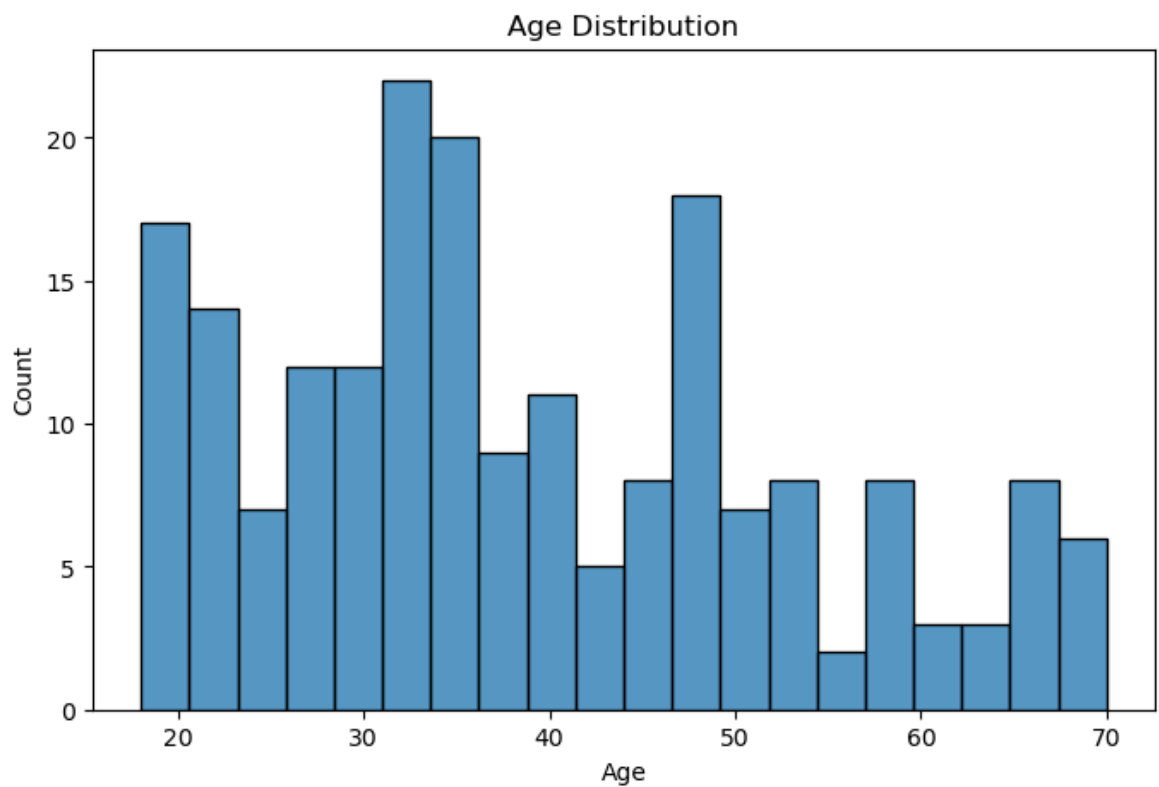
```
Out[10]: np.int64(0)
```

```
In [11]: sns.countplot(x='Gender', data=df)

plt.title("Gender Distribution")
plt.show()
```



```
In [12]: plt.figure(figsize=(8,5))  
sns.histplot(df['Age'], bins=20)  
plt.title("Age Distribution")  
plt.show()
```

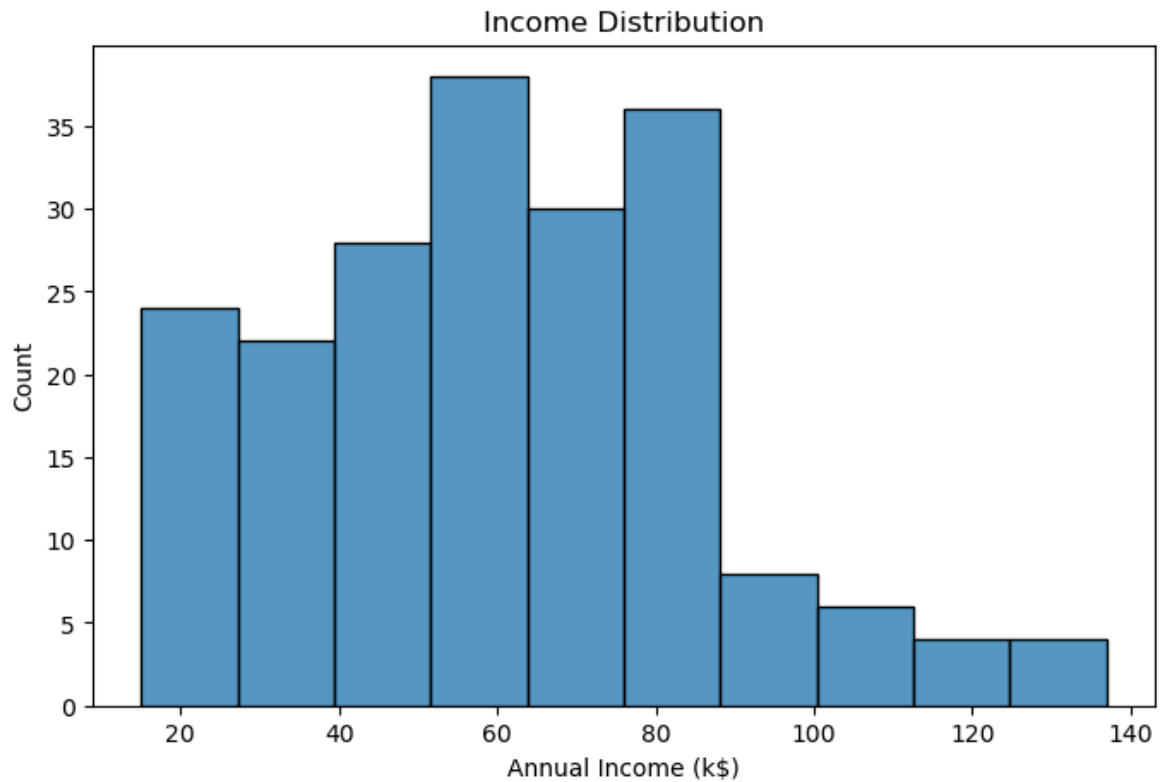


```
In [13]: plt.figure(figsize=(8,5))

sns.histplot(df['Annual Income (k$)'])

plt.title("Income Distribution")

plt.show()
```

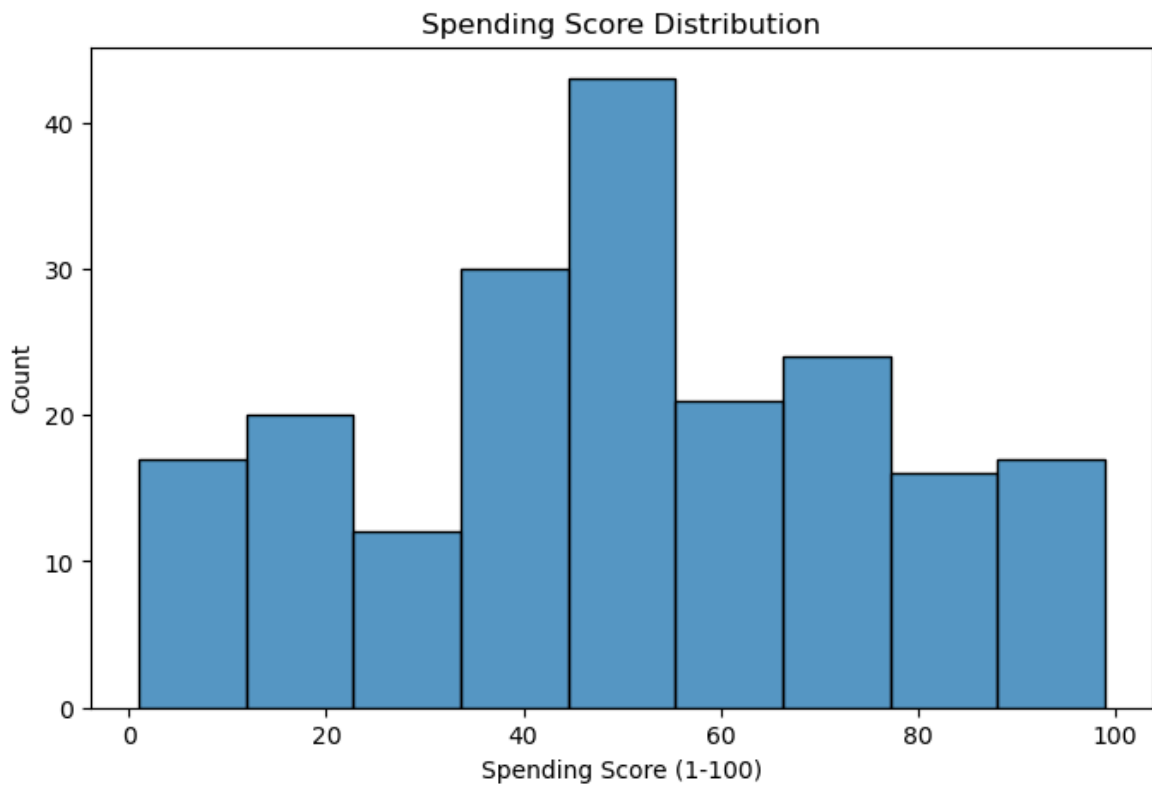


```
In [14]: plt.figure(figsize=(8,5))

sns.histplot(df['Spending Score (1-100)'])

plt.title("Spending Score Distribution")

plt.show()
```



Key Findings from EDA

Gender Distribution

- Female customers slightly outnumber male customers.

Age Distribution

- Most customers belong to the age group of 25–45 years.

Income Distribution

- Customer income is fairly distributed across multiple ranges.

Spending Score Distribution

- Spending behavior varies significantly among customers.

These observations indicate that customer behavior differs substantially and segmentation can provide meaningful insights.

```
In [16]: X = df[['Annual Income (k$)',  
              'Spending Score (1-100)']]
```

```
In [17]: scaler = StandardScaler()  
  
X_scaled = scaler.fit_transform(X)
```

```
In [ ]: wcss = []

for i in range(1,11):

    kmeans = KMeans(
        n_clusters=i,
        random_state=42
    )

    kmeans.fit(X_scaled)

    wcss.append(kmeans.inertia_)
```

```
In [19]: plt.figure(figsize=(8,5))

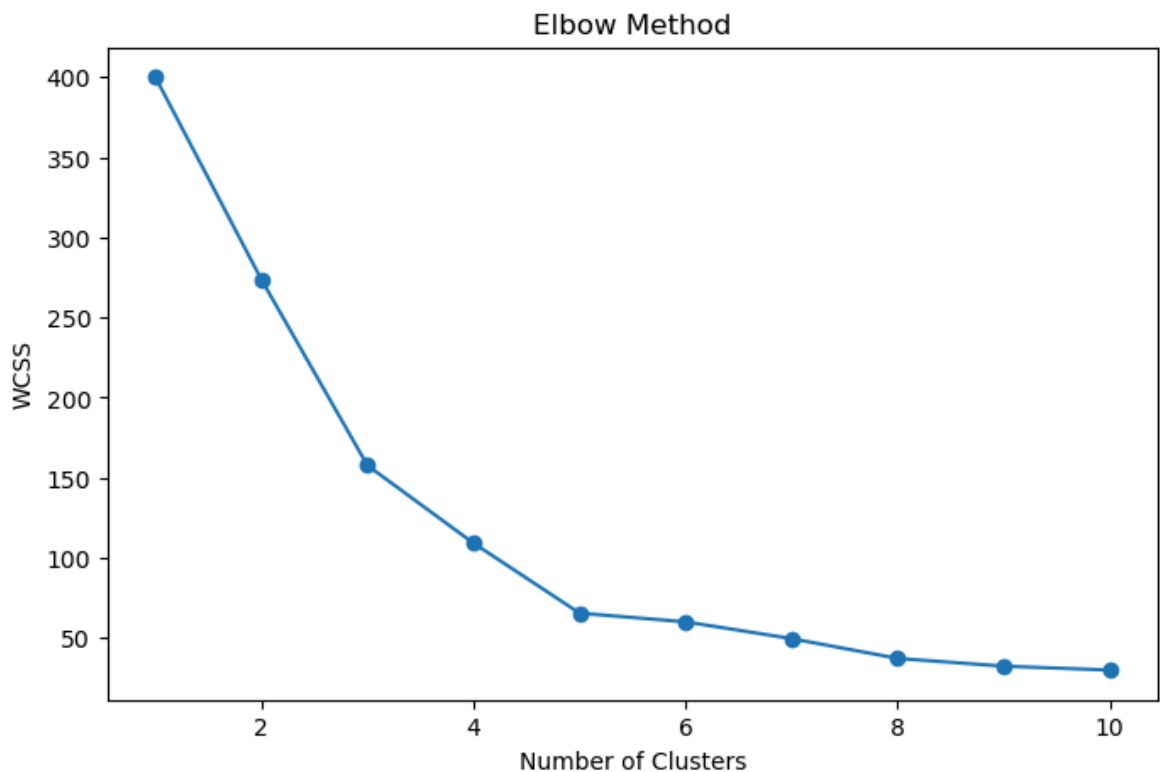
plt.plot(range(1,11), wcss, marker='o')

plt.title("Elbow Method")

plt.xlabel("Number of Clusters")

plt.ylabel("WCSS")

plt.show()
```



Optimal Number of Clusters

The Elbow Method was used to determine the optimal value of K.

Observation: The curve begins to flatten at K = 5.

Therefore:

Optimal Number of Clusters = 5

This value provides a balance between cluster compactness and model simplicity.

```
In [20]: kmeans = KMeans(  
          n_clusters=5,  
          random_state=42  
        )  
  
clusters = kmeans.fit_predict(X_scaled)
```

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
warnings.warn(

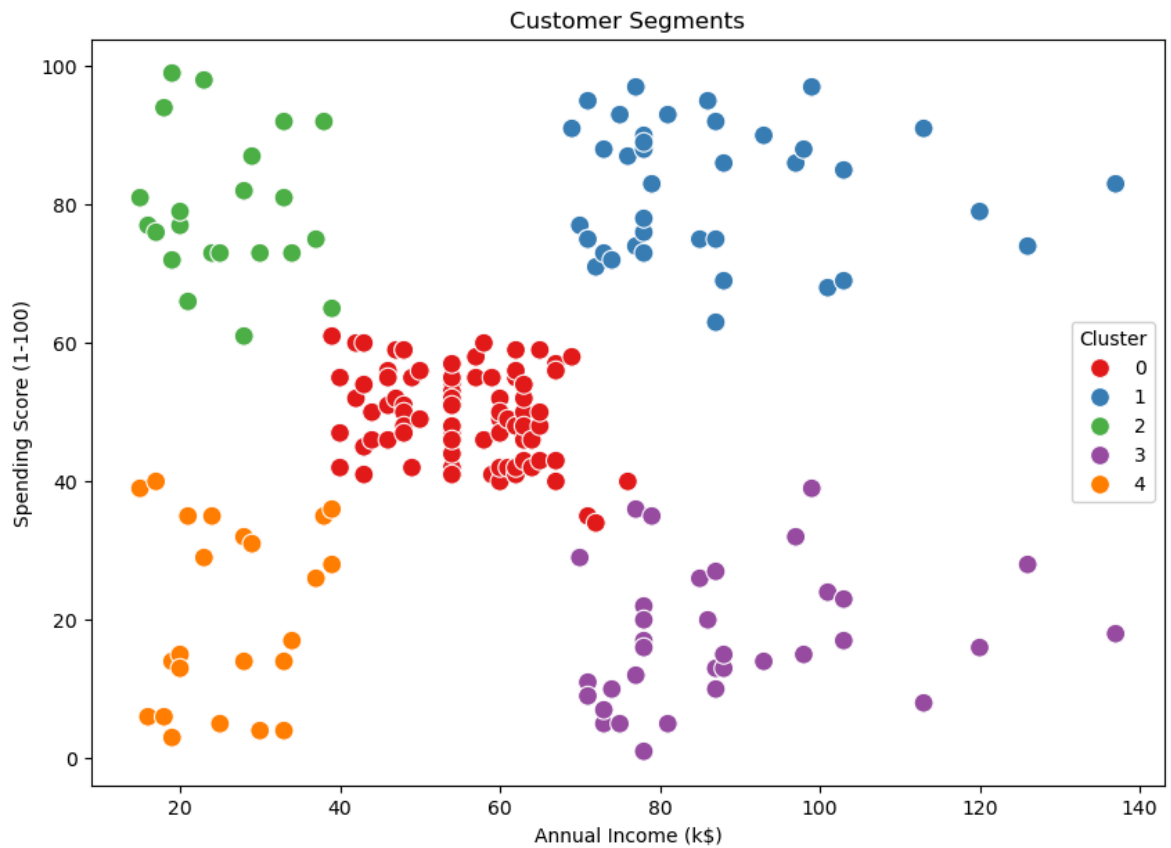
```
In [21]: df['Cluster'] = clusters
```

```
In [22]: df.head()
```

```
Out[22]:
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Cluster
0	1	Male	19	15	39	4
1	2	Male	21	15	81	2
2	3	Female	20	16	6	4
3	4	Female	23	16	77	2
4	5	Female	31	17	40	4

```
In [23]: plt.figure(figsize=(10,7))  
  
sns.scatterplot(  
    x='Annual Income (k$)',  
    y='Spending Score (1-100)',  
    hue='Cluster',  
    palette='Set1',  
    data=df,  
    s=100  
)  
  
plt.title("Customer Segments")  
  
plt.show()
```



Customer Segment Interpretation

Cluster 0

Characteristics:

- Medium Income
- Medium Spending

Business Type: Regular Customers

Recommendation: Maintain engagement through loyalty programs.

Cluster 1

Characteristics:

- High Income
- High Spending

Business Type: Premium Customers

Recommendation: Offer exclusive memberships and premium services.

Cluster 2

Characteristics:

- Low Income
- High Spending

Business Type: Impulse Buyers

Recommendation: Target with promotional offers and discounts.

Cluster 3

Characteristics:

- High Income
- Low Spending

Business Type: Potential Customers

Recommendation: Increase engagement through personalized marketing.

Cluster 4

Characteristics:

- Low Income
- Low Spending

Business Type: Low Priority Customers

Recommendation: Focus on cost-effective marketing strategies.

```
In [25]: cluster_summary = df.groupby('Cluster').mean(numeric_only=True)
cluster_summary
```

```
Out[25]:
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
Cluster				
0	86.320988	42.716049	55.296296	49.518519
1	162.000000	32.692308	86.538462	82.128205
2	23.090909	25.272727	25.727273	79.363636
3	164.371429	41.114286	88.200000	17.114286
4	23.000000	45.217391	26.304348	20.913043

Results

Successfully segmented customers into 5 distinct groups using K-Means Clustering.

Achievements:

- ✓ Data Cleaning Completed
- ✓ Exploratory Data Analysis Performed
- ✓ Feature Scaling Applied
- ✓ Optimal K Identified Using Elbow Method
- ✓ K-Means Clustering Implemented
- ✓ Customer Segments Visualized
- ✓ Business Insights Generated

Conclusion

This project demonstrates how unsupervised machine learning can be used to understand customer behavior.

The K-Means algorithm successfully identified five unique customer segments based on income and spending patterns.

These insights can help businesses:

- Improve marketing efficiency
- Increase customer satisfaction
- Enhance customer retention
- Improve profitability

The project highlights the practical application of clustering techniques in real-world business scenarios.

In []: